



AWS SUMMIT

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DEV-200

Build Agentic Incident Response with Strands, OpenSearch & CloudWatch

Ankit Mehta

Founder, vigileshq.com | AWS Community Builder

AWS is investing heavily in agentic AI.

The building blocks for intelligent incident response are already here.

What if incidents could learn from themselves?

AWS Incident Manager Status

- Deprecated November 2025
- No new customers accepted
- Being sunset with no full replacement
- SRE teams returning to manual workflows

Current Reality

- No native AWS service for end-to-end incident response
- Manual triage and correlation processes
- Alert fatigue from high noise-to-signal ratio
- Every incident starts without historical context

The Missing Intelligence Layer

- Incident knowledge walks out the door with people
- No learning between similar incidents
- Pattern recognition relies entirely on human memory
- Missed opportunity to turn incidents into intelligence

The New Agentic Stack on AWS

Three services. One intelligent system.

Built with what AWS is actively investing in.

Strands Agents SDK : The orchestration layer

- Open source, model-driven orchestration
- Multi-agent workflows with agents-as-tools pattern
- 14M+ downloads, production-ready
- Works natively with AWS Bedrock models

OpenSearch 3.5 : The intelligence brain

- Native MCP integration for agentic workflows
- Hybrid search: neural embeddings + BM25 keywords
- Agentic conversation memory storage
- AI-native features on 2026 roadmap

CloudWatch : The data foundation

- Metrics, alarms, and logs in real-time
- Already deployed in every AWS account
- The data source everyone already has
- No additional infrastructure needed

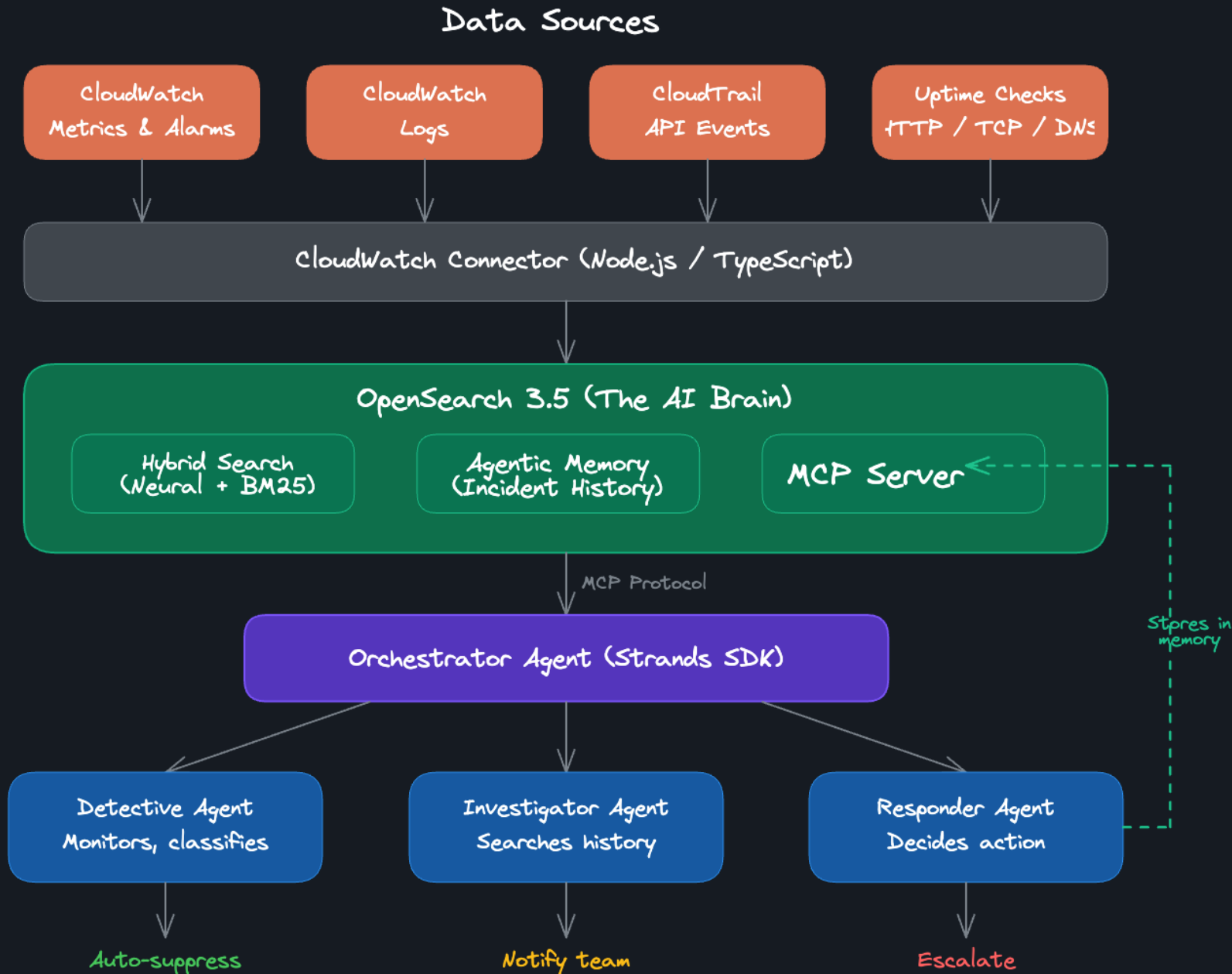
Architecture



Real-time intelligence flow.

From first alert to resolved incident with full context memory.

Built for learning, not just logging.



Intelligent Pattern Recognition

Not just search. True incident intelligence.

The system gets smarter with every resolved incident.

Like having a senior SRE who never forgets.

Hybrid Search in Action

- Query: "API latency spike midnight"
- Neural: Finds similar patterns (cache expiry, batch jobs)
- BM25: Exact alarm name matches
- Result: "90% likely CDN cache refresh - auto-suppress"

Learning from History

- First time: RDS connection exhaustion -> 45min to resolve
- Next time: Same pattern -> Instant recognition
- Root cause: Connection pool leak in service X
- Solution: Restart application, increase pool size
- Auto-escalation: Page database team immediately

Pattern Evolution

- Week 1: Manual investigation of CPU spikes
- Week 4: Auto-classification of batch job patterns
- Month 3: Predictive suppression of known false positives
- Result: 80% reduction in manual triage time

Agents in Action

A real incident response flow.

Three specialized agents
working together.

From detection to resolution in
seconds.

Detective Agent: First Response (0-5 seconds)

- Receives: CloudWatch alarm "RDS-ConnectionCount-High"
- Classifies: Database, Critical severity
- Identifies: Production RDS instance db-prod-01
- Action: Trigger investigation workflow

Investigator Agent: Pattern Analysis (5-15 seconds)

- Searches: Similar "ConnectionCount-High" incidents
- Finds: 3 previous incidents, same service pattern
- Root cause: Connection pool leak in checkout-service
- Recommended action: Restart service + scale RDS

Responder Agent: Decision & Action (15-20 seconds)

- Decision: Auto-escalate (critical + known pattern)
- Actions: Page database team, create incident record
- Memory: Store resolution for future pattern matching
- Outcome: 20-second response vs 45-minute manual

Let's see it in action



The future of incident response is intelligent.

Start building with services AWS is actively investing in.

Your next incident could be your system's first lesson.

The Opportunity is Now

- AWS Incident Manager is deprecated
- Agentic AI building blocks are ready today
- Strands, OpenSearch, Bedrock - all production-ready
- First-mover advantage in intelligent operations

Intelligence Over Automation

- Not just faster alerts - smarter pattern recognition
- System learns from every resolved incident
- Hybrid search finds connections humans miss
- 80% reduction in manual triage time

Start Simple, Scale Smart

- Open source CloudWatch connector available
- 5 lines of code for your first Strands agent
- Build incrementally with existing infrastructure
- Turn your incidents into intelligence



Failure is simply the opportunity to begin again, this time more intelligently.

- Henry Ford



Ankit Mehta Founder, Vigiles
@ankyitm
[linkedin.com/in/ankitmehta-th](https://www.linkedin.com/in/ankitmehta-th)

Cost Savings - OpenSearch

FinOps for Observability: Optimizing OpenSearch Costs Without Sacrificing Visibility

<https://www.youtube.com/watch?v=f4bN96sdUVA>

<https://anky.it/FinOps-Observability-OSC26-China-ankit-mehta.pdf>